

BRAF fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes BRAF gene fusions from the msk_impact_50k_2026 study, covering 179 fusion events. 151/178 fusions (84.8%) were in-frame. 163/178 fusions (91.6%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0133676$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 84.8% in-frame and 91.6% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 179 analyzed fusion events, identifying 67 distinct fusion partner genes. The most frequent partner was KIAA1549, observed in 43 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions retained the domain, $p=0.0133676$ (statistically significant at $\alpha=0.05$). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00699301.

Domain disruption

Disruption of the RAS-binding domain and Cysteine-rich domain was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions disrupted the domain, $p=0.041705$ (statistically significant at $\alpha=0.05$). The RAS-binding domain and Cysteine-rich domain appear to require loss or disruption rather than retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.963504.

Cutpoint detection

Cutpoint detection scanned 178 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 380 aa (exon 8; permutation-corrected $p=0.001998$, statistically significant at $\alpha=0.05$). This is -78 aa from the nearest configured domain boundary at 458 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=163$) and "not_retained" ($n=15$). For Frame_status, retained: 142/152 (93.4%, 95% CI 88.3%-96.4%); not_retained: 9/11 (81.8%, 95% CI 52.3%-94.9%). Welch's t-test comparing Tumor_variant_count between groups gave means of 52.0736 vs 36.4, $p=0.11965$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (67 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for BRAF in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (23 row(s) in genomic_bin_recurrence, 12 row(s) in exact_position_recurrence, 6 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: BRAF has no gene_pair configured; joint-partner analysis was skipped.

Mutation cooccurrence

Mutation cooccurrence produced results for this run (1 row(s) in cooccurrence_results); see the accompanying table.

Window detection

Window detection produced results for this run (57 row(s) in window_scan, 28 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIAA1549	43	0.24022346368715083	0.21553360374650618	0.001614663595055688	0.8847211802950737	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.33366411984501465	1
CUL1	5	0.027932960893854747	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.28565588873695824	2
FAM131B	3	0.01675977653631285	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.28230393342969573	3
SND1	17	0.09497206703910614	0.21553360374650618	0.001614663595055688	0.8317520556609741	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.28214333215548626	4
AKAP9	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2806279557760644	5
CARMIL1	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2806279557760644	6
EXOC4	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2806279557760644	7
MAD1L1	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2806279557760644	8
ABCC1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CAPZA2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	10
CCDC6	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	11
CMAS	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	12
ERC1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	13
KLHL12	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	14
LRBA	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	15
MINDY4	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	16
OSBPL7	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	17
PHTF2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	18
PRIM2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PRKAR1B	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	20
PTPRZ1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	21
RGS3	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	22
SNX8	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	23
TRA2B	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	24
TRIM24	8	0.0446927374301676	0.21553360374650618	0.001614663595055688	0.8951612903225806	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.27657091847204573	25
NRF1	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.9247311827956989	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2709505364212257	26
AGAP3	6	0.0335195530726257	0.21553360374650618	0.001614663595055688	0.8637992831541219	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.26851466208951436	27
GTF2IRD1	3	0.01675977653631285	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.26294909472001826	28
ATF7	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CLEC2L	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	30
DBI	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	31
DOCK4	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	32
GIPC2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	33
GTF2I	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	34
KCND2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	35
NFIA	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	36
OSBPL9	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	37
PAK2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	38
PRKAR2B	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	39

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RRBP1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	40
SETBP1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	41
SLC45A3	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	42
SPOCK1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	43
ZC3HAV1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	44
ZNF207	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	45
CDK5RAP2	4	0.0223463687150838	0.21553360374650618	0.001614663595055688	0.803763440860215	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2561573304381657	46
CLIP1	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.7096774193548387	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2386924719050967	47
AGK	14	0.0782122905027933	0.21553360374650618	0.001614663595055688	0.41167434715821816	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.21410374291917902	48
MKRN1	10	0.055865921787709494	0.21553360374650618	0.001614663595055688	0.4354838709677419	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.21097126087608245	49

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TMPRSS2	4	0.0223463687150838	0.21553360374650618	0.001614663595055688	0.4408602150537635	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.20172184656719797	50
ABCC2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	51
ITGB5	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	52
MKLN1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	53
NEO1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	54
PWWP2A	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	55
TBXAS1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	56
UBE2H	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	57
WEE2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	58
PJA2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3763440860215054	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1870164942514654	59

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SVOPL	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3763440860215054	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1870164942514654	60
EPB41	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18540359102565895	61
PLPP1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18540359102565895	62
RBM33	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18540359102565895	63
RBMS3	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18540359102565895	64
SCRIB	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18540359102565895	65
TRIM33	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18540359102565895	66
LUC7L2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.0	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1305648813482396	67

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
238	1	162	0	15	None	1.0	-0.0
287	1	162	1	14	0.08641975308641975	0.16187392877547135	0.7908230925727642
303	2	161	1	14	0.17391304347826086	0.23330512802756184	0.6320757153442778
327	19	144	6	9	0.19791666666666666	0.00883450422715351	2.053817817119616

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
380	19	144	9	6	0.08796296296296297	4.496509404813937e-05	4.347124493887456
381	114	49	9	6	1.5510204081632653	0.5596655980047007	0.2520713878510084
393	140	23	12	3	1.5217391304347827	0.46313499579777323	0.33429240123968906
400	141	22	12	3	1.6022727272727273	0.44662922460047316	0.3500528628421283
438	141	22	14	1	0.4577922077922078	0.6965277300948016	0.15706158880581256
439	163	0	14	1	None	0.08426966292134831	1.0743287432532127

domain_disruption tables

frame_domain_contingency_table

142	22
9	5

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
ras_binding	RAS-binding domain	178	14	0	164	0.0
cysteine_rich	Cysteine-rich domain	178	14	1	163	0.005699893955461294

domain_retention tables

frame_domain_contingency_table

142	21
9	6

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	178	163	0	15	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	163	179	0.9106145251396648

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ABCC1	1
ABCC2	1
AGAP3	6
AGK	14
AKAP9	2
ATF7	1
CAPZA2	1
CARMIL1	2
CCDC6	1
CDK5RAP2	4
CLEC2L	1
CLIP1	2
CMAS	1
CUL1	5
DBI	1
DOCK4	1
EPB41	1
ERC1	1
EXOC4	2
FAM131B	3
GIPC2	1
GTF2I	1
GTF2IRD1	3
ITGB5	1
KCND2	1
KIAA1549	43
KLHL12	1
LRBA	1
LUC7L2	1
MAD1L1	2
MINDY4	1
MKLN1	1

Partner_gene	Event_count
MKRN1	10
NEO1	1
NFIA	1
NRF1	2
OSBPL7	1
OSBPL9	1
PAK2	1
PHTF2	1
PJA2	1
PLPP1	1
PRIM2	1
PRKAR1B	1
PRKAR2B	1
PTPRZ1	1
PWWP2A	1
RBM33	1
RBMS3	1
RGS3	1
RRBP1	1
SCRIB	1
SETBP1	1
SLC45A3	1
SND1	17
SNX8	1
SPOCK1	1
SVOPL	1
TBXAS1	1
TMPRSS2	4
TRA2B	1
TRIM24	8
TRIM33	1
UBE2H	1
WEE2	1
ZC3HAV1	1
ZNF207	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
7	140493000	140494000	21	EVT-P-0004795-T01-IM5-87, EVT-P-0009024-T01-IM5-190, EVT-P-0009024-T02-IM6-195, EVT-P-0009757-T01-IM5-15, EVT-P-0012395-T01-IM5-217, ... (21 total)
7	140482000	140483000	20	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0007392-T01-IM5-214, EVT-P-0007533-T01-IM5-229, EVT-P-0009246-T01-IM5-103, ... (20 total)
7	140491000	140492000	18	EVT-P-0004203-T01-IM5-116, EVT-P-0005121-T02-IM5-156, EVT-P-0009694-T01-IM5-207, EVT-P-0015985-T03-IM6-111, EVT-P-0021127-T01-IM6-175, ... (18 total)
7	140489000	140490000	14	EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, EVT-P-0008140-T01-IM5-224, EVT-P-0008789-T01-IM5-205, EVT-P-0015540-T01-IM6-163, ... (14 total)
7	140490000	140491000	13	EVT-P-0001687-T01-IM3-220, EVT-P-0006601-T03-IM6-187, EVT-P-0007861-T01-IM5-13, EVT-P-0012575-T01-IM5-151, EVT-P-0014833-T02-IM6-128, ... (13 total)
7	140492000	140493000	12	EVT-P-0014191-T01-IM6-160, EVT-P-0014381-T01-IM6-161, EVT-P-0020125-T01-IM6-177, EVT-P-0027592-T01-IM6-140, EVT-P-0030237-T03-IM6-10, ... (12 total)
7	140487000	140488000	11	EVT-P-0004032-T01-IM5-166, EVT-P-0010494-T01-IM5-90, EVT-P-0024206-T01-IM6-130, EVT-P-0032569-T01-IM6-246, EVT-P-0032569-T02-IM6-247, ... (11 total)
7	140488000	140489000	10	EVT-P-0002197-T01-IM3-228, EVT-P-0005834-T01-IM5-114, EVT-P-0009097-T01-IM5-155, EVT-P-0016557-T01-IM6-172, EVT-P-0024995-T01-IM6-121, ... (10 total)
7	140483000	140484000	9	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0013536-T01-IM5-198, EVT-P-0022778-T01-IM6-237, EVT-P-0023821-T01-IM6-12, ... (9 total)

chromosome	bin_start	bin_end	n_events	event_ids
7	140485000	140486000	7	EVT-P-0008920-T01-IM5-251, EVT-P-0027762-T01-IM6-250, EVT-P-0041308-T02-IM6-213, EVT-P-0044845-T01-IM6-134, EVT-P-0046745-T01-IM6-136, ... (7 total)
7	140486000	140487000	7	EVT-P-0008567-T01-IM5-208, EVT-P-0014377-T01-IM6-99, EVT-P-0016313-T01-IM6-104, EVT-P-0025699-T02-IM6-241, EVT-P-0050196-T01-IM6-126, ... (7 total)
7	140497000	140498000	6	EVT-P-0010945-T01-IM5-26, EVT-P-0015847-T02-IM6-95, EVT-P-0030746-T01-IM6-25, EVT-P-0042835-T02-IM6-8, EVT-P-0042835-T03-IM6-7, ... (6 total)
7	140481000	140482000	5	EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0013777-T01-IM5-158, EVT-P-0021469-T01-IM6-108, EVT-P-0040760-T01-IM6-159
7	140484000	140485000	5	EVT-P-0008083-T02-IM6-197, EVT-P-0008083-T03-IM6-191, EVT-P-0010398-T01-IM5-100, EVT-P-0038194-T01-IM6-115, EVT-P-0054046-T01-IM6-93
7	140495000	140496000	5	EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120, EVT-P-0012884-T01-IM5-210, EVT-P-0024227-T01-IM6-19, EVT-P-0034621-T01-IM6-2
7	140498000	140499000	4	EVT-P-0023378-T01-IM6-21, EVT-P-0025561-T01-IM6-18, EVT-P-0025561-T03-IM6-17, EVT-P-0042835-T03-IM6-109
7	140494000	140495000	3	EVT-P-0001832-T02-IM6-22, EVT-P-0036886-T01-IM6-16, EVT-P-0064223-T01-IM7-107
7	140496000	140497000	2	EVT-P-0000913-T02-IM5-23, EVT-P-0023220-T01-IM6-188
7	140499000	140500000	2	EVT-P-0019039-T01-IM6-248, EVT-P-0050931-T01-IM6-6
7	140434000	140435000	1	EVT-P-0004373-T02-IM5-194
7	140500000	140501000	1	EVT-P-0029531-T01-IM6-24
7	140501000	140502000	1	EVT-P-0006412-T01-IM5-184
7	140502000	140503000	1	EVT-P-0051214-T01-IM6-235

exact_position_recurrence

chromosome	position	n_events	event_ids
7	140482786	2	EVT-P-0028611-T01-IM6-162, EVT-P-0028611-T02-IM6-168

chromosome	position	n_events	event_ids
7	140487437	2	EVT-P-0047918-T02-IM6-132, EVT-P-0047918-T04-IM7-131
7	140487936	2	EVT-P-0032569-T01-IM6-246, EVT-P-0032569-T02-IM6-247
7	140489537	2	EVT-P-0017693-T01-IM6-176, EVT-P-0017693-T02-IM6-149
7	140491093	2	EVT-P-0035153-T01-IM6-144, EVT-P-0035153-T03-IM6-145
7	140491645	2	EVT-P-0030796-T01-IM6-113, EVT-P-0044270-T02-IM6-139
7	140491749	2	EVT-P-0026158-T01-IM6-186, EVT-P-0026158-T04-IM6-185
7	140492893	2	EVT-P-0039648-T01-IM6-122, EVT-P-0039648-T03-IM6-123
7	140493285	2	EVT-P-0035844-T03-IM7-147, EVT-P-0035844-T04-IM7-148
7	140493485	2	EVT-P-0009024-T01-IM5-190, EVT-P-0009024-T02-IM6-195
7	140493530	2	EVT-P-0052411-T03-IM6-154, EVT-P-0052411-T05-IM7-152
7	140497995	2	EVT-P-0042835-T02-IM6-8, EVT-P-0042835-T03-IM6-7

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
381	95	85	6547	False
393	29	29	4129	False
327	22	21	4942	False
439	22	21	1246	False
380	3	3	5397	False
438	2	2	282	False

mutation_cooccurrence tables

cooccurrence_results

comparator_gene	alteration_type	protein_change	label	contingency_table	odds_ratio	p_value	direction	cohort_sample_count	fusion_positive_sample_count	comparator_altered_sample_count	both_altered_sample_count
BRAF	point_mutation	V600E	BRAF point_mutation (V600E)	[2, 175], [1407, 52747]	0.42844552746471726	0.3375200112983454	mutually_exclusive	54331	177	1409	2

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
127	327	200	25	153	19	144	0.19791666666666666	0.00883450422715351	2.053817817119616	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, EVT-P-0007266-T02-IM6-120, ... (25 total)
180	380	200	28	150	19	144	0.08796296296296297	4.496509404813937e-05	4.347124493887456	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0002197-T01-IM3-228, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, ... (28 total)
181	381	200	123	55	114	49	1.5510204081632653	0.5596655980047007	0.2520713878510084	EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (123 total)
193	393	200	152	26	140	23	1.5217391304347827	0.4631349957977323	0.33429240123968906	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (152 total)
200	400	200	153	25	141	22	1.6022727272727273	0.44662922460047316	0.3500528628421283	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (153 total)
227	327	100	25	153	19	144	0.19791666666666666	0.00883450422715351	2.053817817119616	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, EVT-P-0007266-T02-IM6-120, ... (25 total)
238	338	100	25	153	19	144	0.19791666666666666	0.00883450422715351	2.053817817119616	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, EVT-P-0007266-T02-IM6-120, ... (25 total)
238	438	200	155	23	141	22	0.4577922077922078	0.6965277300948016	0.15706158880581256	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (155 total)
277	327	50	24	154	18	145	0.18620689655172415	0.007054608215979324	2.151527100175104	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, EVT-P-0007266-T02-IM6-120, ... (24 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
280	380	100	27	151	18	145	0.08275862068965517	3.162216908790396e-05	4.5000083434297995	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0002197-T01-IM3-228, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, ... (27 total)
281	381	100	122	56	113	50	1.5066666666666666	0.5620256581773301	0.25024385711694297	EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (122 total)
287	337	50	24	154	18	145	0.18620689655172415	0.007054608215979324	2.151527100175104	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, EVT-P-0007266-T02-IM6-120, ... (24 total)
287	387	100	122	56	113	50	1.5066666666666666	0.5620256581773301	0.25024385711694297	EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (122 total)
293	393	100	150	28	139	24	2.106060606060606	0.2608884314370491	0.5835451783717784	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (150 total)
300	400	100	151	27	140	23	2.2134387351778657	0.25011484918241694	0.6018605236771493	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (151 total)
302	327	25	23	155	18	145	0.2482758620689655	0.028754507085544598	1.5412940727493663	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120, EVT-P-0010945-T01-IM5-26, ... (23 total)
303	328	25	23	155	18	145	0.2482758620689655	0.028754507085544598	1.5412940727493663	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120, EVT-P-0010945-T01-IM5-26, ... (23 total)
303	353	50	23	155	18	145	0.2482758620689655	0.028754507085544598	1.5412940727493663	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120, EVT-P-0010945-T01-IM5-26, ... (23 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
303	403	100	151	27	140	23	2.2134387351778657	0.25011484918241694	0.6018605236771493	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (151 total)
327	352	25	22	156	17	146	0.2328767123287671	0.023711801950937986	1.625035441061342	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120, EVT-P-0010945-T01-IM5-26, ... (22 total)
327	377	50	22	156	17	146	0.2328767123287671	0.023711801950937986	1.625035441061342	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120, EVT-P-0010945-T01-IM5-26, ... (22 total)
327	427	100	150	28	139	24	2.106060606060606	0.2608884314370491	0.5835451783717784	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (150 total)
327	527	200	174	4	161	2	12.384615384615385	0.03615501809179204	1.441831416811241	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001036-T01-IM3-192, EVT-P-0001687-T01-IM3-220, ... (174 total)
331	381	50	98	80	95	68	5.588235294117647	0.005720658539439056	2.242553974077208	EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, EVT-P-0002197-T01-IM3-228, EVT-P-0004032-T01-IM5-166, ... (98 total)
338	438	100	130	48	122	41	2.6036585365853657	0.1233591284699147	0.9088287075651992	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (130 total)
339	439	100	152	26	144	19	6.631578947368421	0.0018545198696076845	2.731768509213372	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, ... (152 total)
343	393	50	127	51	121	42	4.321428571428571	0.013112529347571712	1.8823135267889146	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (127 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
350	400	50	128	50	122	41	4.463414634146342	0.01259693916356655	1.8997349680458389	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (128 total)
356	381	25	98	80	95	68	5.588235294117647	0.005720658539439056	2.242553974077208	EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, EVT-P-0002197-T01-IM3-228, EVT-P-0004032-T01-IM5-166, ... (98 total)
368	393	25	127	51	121	42	4.321428571428571	0.013112529347571712	1.8823135267889146	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (127 total)
375	400	25	128	50	122	41	4.463414634146342	0.01259693916356655	1.8997349680458389	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (128 total)
380	405	25	128	50	122	41	4.463414634146342	0.01259693916356655	1.8997349680458389	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (128 total)
380	430	50	128	50	122	41	4.463414634146342	0.01259693916356655	1.8997349680458389	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (128 total)
380	480	100	152	26	144	19	6.631578947368421	0.0018545198696076845	2.731768509213372	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, ... (152 total)
380	580	200	152	26	144	19	6.631578947368421	0.0018545198696076845	2.731768509213372	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, ... (152 total)
381	406	25	125	53	122	41	11.902439024390244	3.8505600666833e-05	4.414476097457752	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (125 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
381	431	50	125	53	122	41	11.902439024390244	3.8505600666833e-05	4.414476097457752	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (125 total)
381	481	100	149	29	144	19	15.157894736842104	5.1142248335696075e-06	5.291220183264657	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, ... (149 total)
381	581	200	149	29	144	19	15.157894736842104	5.1142248335696075e-06	5.291220183264657	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, ... (149 total)
388	438	50	32	146	27	136	0.39705882352941174	0.15129618600521133	0.8201720198801077	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0003502-T01-IM5-101, EVT-P-0008083-T02-IM6-197, EVT-P-0008083-T03-IM6-191, ... (32 total)
389	439	50	54	124	49	114	0.8596491228070176	0.7751756210061139	0.11059989414744384	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, ... (54 total)
393	418	25	30	148	27	136	0.7941176470588235	0.72098041318196	0.14207653356425853	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0008083-T02-IM6-197, EVT-P-0008083-T03-IM6-191, EVT-P-0008567-T01-IM5-208, ... (30 total)
393	443	50	54	124	49	114	0.8596491228070176	0.7751756210061139	0.11059989414744384	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, ... (54 total)
393	493	100	54	124	49	114	0.8596491228070176	0.7751756210061139	0.11059989414744384	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, ... (54 total)
393	593	200	54	124	49	114	0.8596491228070176	0.7751756210061139	0.11059989414744384	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, ... (54 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
400	450	50	25	153	23	140	1.0678571428571428	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (25 total)
400	500	100	25	153	23	140	1.0678571428571428	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (25 total)
400	600	200	25	153	23	140	1.0678571428571428	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (25 total)
414	439	25	24	154	22	141	1.0141843971631206	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (24 total)
438	463	25	24	154	22	141	1.0141843971631206	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (24 total)
438	488	50	24	154	22	141	1.0141843971631206	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (24 total)
438	538	100	24	154	22	141	1.0141843971631206	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (24 total)
438	638	200	24	154	22	141	1.0141843971631206	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (24 total)
439	464	25	22	156	22	141	None	0.22175613072495187	0.6541243647173782	EVT-P-0001036-T01-IM3-192, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, EVT-P-0007533-T01-IM5-229, ... (22 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
439	489	50	22	156	22	141	None	0.22175613072495187	0.6541243647173782	EVT-P-0001036-T01-IM3-192, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, EVT-P-0007533-T01-IM5-229, ... (22 total)
439	539	100	22	156	22	141	None	0.22175613072495187	0.6541243647173782	EVT-P-0001036-T01-IM3-192, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, EVT-P-0007533-T01-IM5-229, ... (22 total)
439	639	200	22	156	22	141	None	0.22175613072495187	0.6541243647173782	EVT-P-0001036-T01-IM3-192, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, EVT-P-0007533-T01-IM5-229, ... (22 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
381	481	100	149	29	144	19	15.157894736842104	5.1142248335696075e-06	5.291220183264657	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, ... (149 total)
280	380	100	27	151	18	145	0.08275862068965517	3.162216908790396e-05	4.5000083434297995	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0002197-T01-IM3-228, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, ... (27 total)
381	406	25	125	53	122	41	11.902439024390244	3.8505600666833e-05	4.414476097457752	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (125 total)
180	380	200	28	150	19	144	0.08796296296296297	4.496509404813937e-05	4.347124493887456	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0002197-T01-IM3-228, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, ... (28 total)
339	439	100	152	26	144	19	6.631578947368421	0.0018545198696076845	2.731768509213372	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, ... (152 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
356	381	25	98	80	95	68	5.588235294117647	0.005720658539439056	2.242553974077208	EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, EVT-P-0002197-T01-IM3-228, EVT-P-0004032-T01-IM5-166, ... (98 total)
277	327	50	24	154	18	145	0.18620689655172415	0.007054608215979324	2.151527100175104	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, EVT-P-0007266-T02-IM6-120, ... (24 total)
227	327	100	25	153	19	144	0.19791666666666666	0.00883450422715351	2.053817817119616	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, EVT-P-0007266-T02-IM6-120, ... (25 total)
375	400	25	128	50	122	41	4.463414634146342	0.01259693916356655	1.8997349680458389	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (128 total)
368	393	25	127	51	121	42	4.321428571428571	0.013112529347571712	1.8823135267889146	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (127 total)
327	352	25	22	156	17	146	0.2328767123287671	0.023711801950937986	1.625035441061342	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120, EVT-P-0010945-T01-IM5-26, ... (22 total)
302	327	25	23	155	18	145	0.2482758620689655	0.028754507085544598	1.5412940727493663	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120, EVT-P-0010945-T01-IM5-26, ... (23 total)
327	527	200	174	4	161	2	12.384615384615385	0.03615501809179204	1.441831416811241	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001036-T01-IM3-192, EVT-P-0001687-T01-IM3-220, ... (174 total)
338	438	100	130	48	122	41	2.6036585365853657	0.1233591284699147	0.9088287075651992	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (130 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
388	438	50	32	146	27	136	0.39705882352941174	0.15129618600521133	0.8201720198801077	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0003502-T01-IM5-101, EVT-P-0008083-T02-IM6-197, EVT-P-0008083-T03-IM6-191, ... (32 total)
439	464	25	22	156	22	141	None	0.22175613072495187	0.6541243647173782	EVT-P-0001036-T01-IM3-192, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, EVT-P-0007533-T01-IM5-229, ... (22 total)
300	400	100	151	27	140	23	2.2134387351778657	0.25011484918241694	0.6018605236771493	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (151 total)
293	393	100	150	28	139	24	2.106060606060606	0.2608884314370491	0.5835451783717784	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (150 total)
327	427	100	150	28	139	24	2.106060606060606	0.2608884314370491	0.5835451783717784	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (150 total)
200	400	200	153	25	141	22	1.6022727272727273	0.44662922460047316	0.3500528628421283	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (153 total)
193	393	200	152	26	140	23	1.5217391304347827	0.46313499579777323	0.33429240123968906	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (152 total)
181	381	200	123	55	114	49	1.5510204081632653	0.5596655980047007	0.2520713878510084	EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (123 total)
281	381	100	122	56	113	50	1.5066666666666666	0.5620256581773301	0.25024385711694297	EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (122 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
238	438	200	155	23	141	22	0.4577922077922078	0.6965277300948016	0.15706158880581256	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (155 total)
393	418	25	30	148	27	136	0.7941176470588235	0.72098041318196	0.14207653356425853	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0008083-T02-IM6-197, EVT-P-0008083-T03-IM6-191, EVT-P-0008567-T01-IM5-208, ... (30 total)
389	439	50	54	124	49	114	0.8596491228070176	0.7751756210061139	0.11059989414744384	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, ... (54 total)
414	439	25	24	154	22	141	1.0141843971631206	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (24 total)
400	450	50	25	153	23	140	1.0678571428571428	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (25 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021458-T01-IM6-1	P-0021458-T01-IM6		Melanoma	SKCM	ABCC1-BRAF	ABCC1	in-frame	three_prime	140493152	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC1 (NM_004996) - BRAF (NM_004333) rearrangement: t(7;16)(q34;p13.11)(chr7:g.140493152::chr16:g.16116105) Protein Fusion: in frame {ABCC1: BRAF}	NA
EVT-P-0034621-T01-IM6-2	P-0034621-T01-IM6		Melanoma	SKCM	ABCC2-BRAF	ABCC2	unknown	three_prime	140495500	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC2 (NM_000392) - BRAF (NM_004333) rearrangement: t(7;10)(q34;24.2)(chr7:g.140495500::chr10:g.101559038) Protein Fusion: mid-exon {ABCC2: BRAF}	NA
EVT-P-0052582-T01-IM6-3	P-0052582-T01-IM6		Gastrointestinal Stromal Tumor	GIST	AGAP3-BRAF	AGAP3	unknown	three_prime	140485658	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1637:AGAP3_c.1177+1690: BRAFInv Protein Fusion: mid-exon {AGAP3: BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0050574-T02-IM6-4	P-0050574-T02-IM6		Colorectal Cancer	COAD	AGAP3-BRAF	AGAP3	in-frame	three_prime	140493208	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c. 1222-2059:AGAP3_c.1140+900: BRAFInv Protein Fusion: in frame {AGAP3: BRAF}	NA
EVT-P-0039799-T01-IM6-5	P-0039799-T01-IM6		Ovarian Cancer	LGSOC	AGAP3-BRAF	AGAP3	in-frame	three_prime	140482317	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c. 1221+416:AGAP3_c.1314+504: BRAFInv Protein Fusion: in frame {AGAP3: BRAF}	NA
EVT-P-0050931-T01-IM6-6	P-0050931-T01-IM6		Melanoma	SKCM	AGK-BRAF	AGK	in-frame	three_prime	140499251	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c. 102-14294:AGK_c.980+911: BRAFInv Protein Fusion: in frame {AGK: BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042835-T03-IM6-7	P-0042835-T03-IM6		Glioma	GB	AGK-BRAF	AGK	in-frame	three_prime	140497995	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c. 102-13644:AGK_c.980+2167:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0042835-T02-IM6-8	P-0042835-T02-IM6		Glioma	GB	AGK-BRAF	AGK	in-frame	three_prime	140497995	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c. 102-13644:AGK_c.980+2167:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0050635-T01-IM6-9	P-0050635-T01-IM6		Breast Cancer	IDC	AKAP9-BRAF	AKAP9	in-frame	three_prime	140487501	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c. 6613-1489:AKAP9_c.1141-117:BRRAFInv Protein Fusion: in frame {AKAP9:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030237-T03-IM6-10	P-0030237-T03-IM6		Melanoma	MEL	AKAP9-BRAF	AKAP9	in-frame	three_prime	140492357	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.3318+1378:AKAP9_c.1140+1751: BRAFInv Protein Fusion: in frame {AKAP9: BRAF}	NA
EVT-P-0023821-T01-IM6-12	P-0023821-T01-IM6		Melanoma	SKCM	ATF7-BRAF	ATF7	in-frame	three_prime	140483233	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ATF7 (NM_006856) - BRAF (NM_004333) Rearrangement: t(7;12)(q34;q13.13)(c17:g.140483233::chr12:g.53913299) Protein Fusion: in frame {ATF7: BRAF}	NA
EVT-P-0007861-T01-IM5-13	P-0007861-T01-IM5		Non-Small Cell Lung Cancer	LUAD	BRAF-A GAP3	AGAP3	in-frame	three_prime	140490697	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1326+220:AGAP3_c.1141-3313: BRAF inv Protein fusion: in frame (AGAP3-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021519-T02-IM6-14	P-0021519-T02-IM6		Colorectal Cancer	COAD	BRAF-AGAP3	AGAP3	in-frame	three_prime	140493112	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) Rearrangement : c.1326+142:AGAP3_c.1140+996:BRAFinv Protein Fusion: in frame {AGAP3: BRAF}	NA
EVT-P-0009757-T01-IM5-15	P-0009757-T01-IM5		Colorectal Cancer	COAD	BRAF-AGAP3	AGAP3	in-frame	three_prime	140493871	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement : c.1222-2154:AGAP3_c.1140+237:BRAFinv Protein fusion: in frame (AGAP3-BRAF)	NA
EVT-P-0036886-T01-IM6-16	P-0036886-T01-IM6		Ovarian Cancer	HGSOC	BRAF-AGK	AGK	in-frame	three_prime	140494309	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+10859:AGK_c.981-42:BRAFinv Protein Fusion: in frame {AGK:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025561-T03-IM6-17	P-0025561-T03-IM6		Melanoma	SKCM	BRAF-AGK	AGK	in-frame	three_prime	140498719	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 8-18): c. 980+1288:BRAF_c.101+772:AGKinv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0025561-T01-IM6-18	P-0025561-T01-IM6		Melanoma	SKCM	BRAF-AGK	AGK	in-frame	five_prime	140498874	7	327	True	326	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) Rearrangement: c.980+1288:BRAF_c.101+772:AGKinv Protein Fusion: in frame {BRAF:AGK}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024227-T01-IM6-19	P-0024227-T01-IM6		Glioma	GB	BRAF-AGK	AGK	in-frame	three_prime	140495070	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+16796:AGK_c.981-803:BRAFinv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0004477-T01-IM5-20	P-0004477-T01-IM5		Breast Cancer	ILC	BRAF-AGK	AGK	in-frame	three_prime	140495405	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1 to 2 and BRAF exons 8 to 18): c.101+10076:AGK_c.981-1138:BRAFinv Protein fusion: in frame (AGK-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023378-T01-IM6-21	P-0023378-T01-IM6		Melanoma	SKCM	BRAF-AGK	AGK	in-frame	five_prime	140498362	7	327	True	326	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) rearrangement: c.980+1800: BRAF_c.101+14574:AGKinv Protein Fusion: in frame {BRAF:AGK}	NA
EVT-P-0001832-T02-IM6-22	P-0001832-T02-IM6		Non-Small Cell Lung Cancer	LUAD	BRAF-AGK	AGK	in-frame	three_prime	140494489	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exon 1 fused in-frame to BRAF exons 8-18): c.102-572: AGK_c.981-222: BRAFinv Protein fusion: in frame (AGK-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000913-T02-IM5-23	P-0000913-T02-IM5		Non-Small Cell Lung Cancer	LUAD	BRAF-AGK	AGK	in-frame	three_prime	140496386	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK_c.981-2119:BRAFInv Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0029531-T01-IM6-24	P-0029531-T01-IM6		Glioma	GB	BRAF-AGK	AGK	unknown	three_prime	140500233	7	303	False	303	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 7-18): c.101+4572:AGK_c.909:BRAFInv Protein Fusion: mid-exon {AGK: BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030746-T01-IM6-25	P-0030746-T01-IM6		Glioma	GNG	BRAF-AGK	AGK	in-frame	three_prime	140497533	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.101+8348:AGK_c.980+2629:BRAF inv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0010945-T01-IM5-26	P-0010945-T01-IM5		Non-Small Cell Lung Cancer	LUAD	BRAF-AGK	AGK	in-frame	three_prime	140497610	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAF inv Protein fusion: in frame (AGK-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049035-T01-IM6-85	P-0049035-T01-IM6		Glioma	GNG	BRAF-CLEC2L	CLEC2L	in-frame	five_prime	140485799	9	393	True	392	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CLEC2L (NM_001080511) rearrangement: c.1177+1549:BR AF_c.191-5932: CLEC2Linv Protein Fusion: in frame {BRAF:CLEC2L}	NA
EVT-P-004795-T01-IM5-87	P-004795-T01-IM5		Colorectal Cancer	COAD	BRAF-CUL1	CUL1	in-frame	three_prime	140493811	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) Rearrangement: c.790-427: CUL1_c.1140+297: BRAFInv Protein fusion: in frame (CUL1-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022257-T01-IM6-88	P-0022257-T01-IM6		Thyroid Cancer	THME	BRAF-EPB41	EPB41	in-frame	three_prime	140482687	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EPB41 (NM_001166005) - BRAF (NM_004333) fusion (EPB41 exons 1-5 fused in-frame with BRAF exons 11-18): t(1;7)(p35.4;q34)(chr1:g.29424227::chr7:g.140482687) Protein Fusion: in frame {EPB41: BRAF}	NA
EVT-P-0001988-T01-IM3-89	P-0001988-T01-IM3		Glioma	PAST	BRAF-FAM131B	FAM131B	in-frame	three_prime	140489629	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) Deletion : c.175-103:FAM131B_c.1141-2245:BRAFdel Protein fusion: in frame (FAM131B-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0010494-T01-IM5-90	P-0010494-T01-IM5		Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655) -BRAF (NM_004333) Rearrangement : t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) Protein fusion: in frame (GIPC2-BRAF)	NA
EVT-P-0054046-T01-IM6-93	P-0054046-T01-IM6		Colorectal Cancer	COAD	BRAF-GTF2IRD1	GTF2IRD1	in-frame	five_prime	140484706	9	393	True	392	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-1749:BRRAF_c.2320+1046:GTF2IRD1inv Protein Fusion: in frame {BRAF:GTF2IRD1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000828-T03-IM6-94	P-000828-T03-IM6		Prostate Cancer	PRAD	BRAF-GTF2IRD1	GTF2IRD1	in-frame	five_prime	140483554	9	393	True	392	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-597:BRAF_c.2320+14181:GTF2IRD1inv Protein Fusion: in frame {BRAF:GTF2IRD1}	NA
EVT-P-015847-T02-IM6-95	P-0015847-T02-IM6		Soft Tissue Sarcoma	SARCNOS	BRAF-ITGB5	ITGB5	out-of-frame	five_prime	140497019	7	327	True	326	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - ITGB5 (NM_002213) Rearrangement: t(3;7)(q21.2;q34)(c17:3:g.124579522::chr7:g.140497019) Protein Fusion: out of frame {BRAF:ITGB5}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014377-T01-IM6-99	P-0014377-T01-IM6		Melanoma	ACRM	BRAF-NFIA	NFIA	in-frame	three_prime	140486673	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NFIA (NM_001145512) - BRAF (NM_004333) rearrangement: t(1;7)(p31.3;q34)(chr1:g.61836194::chr7:g.140486673) Protein fusion: in frame (NFIA-BRAF)	NA
EVT-P-0010398-T01-IM5-100	P-0010398-T01-IM5		Soft Tissue Sarcoma	MFH	BRAF-OSBPL9	OSBPL9	in-frame	three_prime	140484912	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		OSBPL9 (NM_148909) - BRAF (NM_004333) Rearrangement: t(1;7)(p32.3;q34)(chr1:g.52228623::chr7:g.140484912) Protein fusion: in frame (OSBPL9-BRAF)	NA
EVT-P-0003502-T01-IM5-101	P-0003502-T01-IM5		Non-Small Cell Lung Cancer	LUAD	BRAF-PJA2	PJA2	in-frame	five_prime	140482379	10	438	True	438	True	exact match	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA Protein fusion: in frame (BRAF-PJA2)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036870-T01-IM6-102	P-0036870-T01-IM6		Prostate Cancer	PRAD	BRAF-PRIM2	PRIM2	in-frame	three_prime	140493232	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRIM2 (NM_000947) - BRAF (NM_004333) Rearrangement : t(6;7)(p11.1;q34)(chr6:g.57410815::chr7:g.140493232) Protein Fusion: in frame {PRIM2: BRAF}	NA
EVT-P-0009246-T01-IM5-103	P-0009246-T01-IM5		Soft Tissue Sarcoma	DDLS	BRAF-RBM33	RBM33	in-frame	three_prime	140482661	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RBM33 (NM_053043) - BRAF (NM_004333) rearrangement: c.171+526:RBM33_c.1314+160:BRAFinv Protein fusion: in frame (RBM33-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0016313-T01-IM6-104	P-0016313-T01-IM6		Prostate Cancer	PRAD	BRAF-SLC45A3	SLC45A3	unknown	three_prime	140486608	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SLC45A3 (NM_033102) - BRAF (NM_004333) fusion (SLC45A3 exons 1 fused to BRAF exons 10-18); t(1;7)(q32.1;q34)(c hr1:g.205638452::chr7:g.140486608) Transcript Fusion { SLC45A3:BRAF}	NA
EVT-P-0064764-T01-IM7-105	P-0064764-T01-IM7		Pancreatic Cancer	PAAC	BRAF-SND1	SND1	in-frame	five_prime	140493289	8	380	True	380	True	exact match	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - SND1 (NM_014390) fusion: c.1140+819:BRAF_c.1039-3950:SND1inv Protein Fusion: in frame {BRAF:SND1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029981-T01-IM6-106	P-0029981-T01-IM6		Breast Cancer	BRCA	BRAF-SPOCK1	SPOCK1	in-frame	three_prime	140483862	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SPOCK1 (NM_004598) - BRAF (NM_004333) fusion: t(5;7)(q31.2;q34)(chr5:g.136401840::chr7:g.140483862) Protein Fusion: in frame {SPOCK1:BRAF}	NA
EVT-P-0064223-T01-IM7-107	P-0064223-T01-IM7		Endometrial Cancer	USC	BRAF-TBXAS1	TBXAS1	out-of-frame	five_prime	140494512	7	327	True	326	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - TBXAS1 (NM_001061) rearrangement: c.981-245: BRAF_c.336+2985:TBXAS1inv Protein Fusion: out of frame {BRAF:TBXAS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021469-T01-IM6-108	P-0021469-T01-IM6		Melanoma	SKCM	BRAF-TRIM33	TRIM33	in-frame	three_prime	140481644	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM33 (NM_015906) - BRAF(NM_004333) rearrangement: t(1;7)(p13.2;q34)(chr1:g.114963985::chr7:g.140481644) Protein Fusion: in frame {TRIM33:BRAF}	NA
EVT-P-0042835-T03-IM6-109	P-0042835-T03-IM6		Glioma	GB	BRAF-WEE2	WEE2	unknown	five_prime	140498912	7	327	True	326	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - WEE2 (NM_001105558) rearrangement: c.980+1250:BRAF_c.1305:WEE2inv Protein Fusion: mid-exon {BRAF:WEE2}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015985-T03-IM6-111	P-0015985-T03-IM6		Glioma	ASTR	CAPZA2-BRAF	CAPZA2	in-frame	three_prime	140491425	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CAPZA2 (NM_006136) - BRAF (NM_004333) fusion: c.40-938: CAPZA2_c.1140+2683: BRAF Protein Fusion: in frame {CAPZA2:BRAF}	NA
EVT-P-0033764-T04-IM6-112	P-0033764-T04-IM6		Non-Small Cell Lung Cancer	LUAD	CARMIL1-BRAF	CARMIL1	in-frame	three_prime	140491509	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRRC16A (NM_017640) - BRAF (NM_004333) fusion: t(6;7)(p22.2;q34)(chr6:g.25554753::chr7:g.140491509) Protein Fusion: in frame {LRRC16A:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030796-T01-IM6-113	P-0030796-T01-IM6		Melanoma	ACRM	CARMIL1-BRAF	CARMIL1	in-frame	three_prime	140491645	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRRC16A (NM_017640) - BRAF (NM_004333) fusion (LRRC16A exons 1-29 fused to BRAF exons 9-18): t(6;7)(p22.2;q34)(chr6:g.25580104::chr7:g.140491645) Protein Fusion: in frame {LRRC16A:BRAF}	NA
EVT-P-0005834-T01-IM5-114	P-0005834-T01-IM5		Glioma	PXA	CCDC6-BRAF	CCDC6	in-frame	three_prime	140488401	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CCDC6 (NM_005436) -BRAf (NM_004333) Rearrangement : t(10;7)(chr10q21.2;q34)(chr10:g.61627617::chr7:g.140488401) Protein fusion: in frame (CCDC6-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038194-T01-IM6-115	P-0038194-T01-IM6		Thyroid Cancer	THPA	CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140484954	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(q34;q33.2)(chr7:g.140484954::chr9:g.123187156) Protein Fusion: in frame {CDK5RAP2:BRAF}	NA
EVT-P-0004203-T01-IM5-116	P-0004203-T01-IM5		Melanoma	SKCM	CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140491585	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(q34;q33.2)(chr7:140491585::chr9:123251876) Protein fusion: in frame (CDK5RAP2-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005461-T02-IM5-117	P-0005461-T02-IM5		Melanoma	SKCM	CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140481684	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement : t(7;9)(7q34;9q33.2)(chr7:140481684::chr9:123255452) Protein fusion: in frame (CDK5RAP2-BRAF)	NA
EVT-P-0014969-T01-IM6-118	P-0014969-T01-IM6		Melanoma	MUP	CDK5RAP2-BRAF	CDK5RAP2	in-frame	five_prime	140493700	8	380	True	380	True	exact match	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CDK5RAP2 (NM_018249) Rearrangement : t(7;9)(q34;q33.3)(chr7:g.140493700::chr9:g.123269531) Protein fusion: in frame (BRAF-CDK5RAP2)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040461-T01-IM6-119	P-0040461-T01-IM6		Thyroid Cancer	THPA	CLIP1-BRAF	CLIP1	in-frame	three_prime	140488218	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140488218::chr12:g.122833264) Protein Fusion: in frame {CLIP1:BRAF}	NA
EVT-P-007266-T02-IM6-120	P-007266-T02-IM6		Non-Small Cell Lung Cancer	LUAD	CLIP1-BRAF	CLIP1	in-frame	three_prime	140495673	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140495673::chr12:g.122767026) Protein Fusion: in frame {CLIP1:BRAF}	NA
EVT-P-0024995-T01-IM6-121	P-0024995-T01-IM6		Melanoma	SKCM	CMAS-BRAF	CMAS	unknown	three_prime	140488046	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) rearrangement: t(7;12)(q34;p12.1)(chr7:g.140488046::chr12:g.22218755) Transcript Fusion {CMAS:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039648-T01-IM6-122	P-0039648-T01-IM6		Cancer of Unknown Primary	CUPNOS	CUL1-BRAF	CUL1	in-frame	three_prime	140492893	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215:BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-0039648-T03-IM6-123	P-0039648-T03-IM6		Prostate Cancer	PRAD	CUL1-BRAF	CUL1	in-frame	three_prime	140492893	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215:BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-0064088-T01-IM7-124	P-0064088-T01-IM7		Thyroid Cancer	THAP	CUL1-BRAF	CUL1	in-frame	three_prime	140489451	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+1956:CUL1_c.1141-2067:BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068173-T01-IM7-125	P-0068173-T01-I M7		Melanoma	SKCM	CUL1-BRAF	CUL1	in-frame	three_prime	14049190	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+191:CUL1_c.1140+2118:BRAF Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-0050196-T01-IM6-126	P-0050196-T01-I M6		Prostate Cancer	PRAD	DBI-BRAF	DBI	in-frame	three_prime	140486284	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		DBI (NM_001178017) - BRAF (NM_004333) fusion: t(2;7)(q14.2;q34)(c12hr2:g.120128894::chr7:g.140486284) Protein Fusion: in frame {DBI:BRAF}	NA
EVT-P-0064588-T01-IM7-127	P-0064588-T01-I M7		Prostate Cancer	PRAD	DOCK4-BRAF	DOCK4	out-of-frame	three_prime	140483872	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		DOCK4 (NM_014705) - BRAF (NM_004333) fusion: c.4650+298:DOCK4_c.1178-915:BRAFdup Protein Fusion: out of frame {DOCK4:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014833-T02-IM6-128	P-0014833-T02-IM6		Non-Small Cell Lung Cancer	LUAD	ERC1-BRAF	ERC1	in-frame	three_prime	140490668	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ERC1 (NM_178040) - BRAF (NM_004333) fusion: t(7;12)(q34;p13.33)(chr7:g.140490668::chr12:g.1246645) Protein Fusion: in frame {ERC1:BRAF}	NA
EVT-P-0036937-T01-IM6-129	P-0036937-T01-IM6		Thyroid Cancer	THPA	EXOC4-BRAF	EXOC4	out-of-frame	three_prime	140490241	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) fusion: c.1514+93469:EXOC4_c.1141-2857: BRAFInv Protein Fusion: out of frame {EXOC4:BRAF}	NA
EVT-P-0024206-T01-IM6-130	P-0024206-T01-IM6		Breast Cancer	IDC	EXOC4-BRAF	EXOC4	out-of-frame	three_prime	140487956	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) rearrangement: c.87-8072:EXOC4_c.1141-572: BRAFInv Protein Fusion: out of frame {EXOC4:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047918-T04-IM7-131	P-0047918-T04-IM7		Glioma	PAST	FAM131B-BRAF	FAM131B	in-frame	three_prime	140487437	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_0103169.0) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAF del Protein Fusion: in frame {FAM131B:BRAF}	NA
EVT-P-0047918-T02-IM6-132	P-0047918-T02-IM6		Glioma	PAST	FAM131B-BRAF	FAM131B	in-frame	three_prime	140487437	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_0103169.0) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAF del Protein Fusion: in frame {FAM131B:BRAF}	NA
EVT-P-0044845-T01-IM6-134	P-0044845-T01-IM6		Melanoma	ACRM	GTF2I-BRAF	GTF2I	unknown	three_prime	140485631	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2I (NM_032999) - BRAF (NM_004333) fusion: c.768:GTF2I_c.1177+1717:BRAFInv Protein Fusion: mid-exon {GTF2I:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000828-T05-IM6-135	P-0000828-T05-IM6		Prostate Cancer	PRAD	GTF2IRD1-BRAF	GTF2IRD1	in-frame	three_prime	140483546	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2IRD1 (NM_005685) - BRAF (NM_004333) rearrangement: c.2320+14147:GTF2IRD1_c.1178-589:BRAFinv Protein Fusion: in frame {GTF2IRD1:BRAF}	NA
EVT-P-0046745-T01-IM6-136	P-0046745-T01-IM6		Ovarian Cancer	HGSOC	KCND2-BRAF	KCND2	out-of-frame	three_prime	140485867	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KCND2 (NM_012281) - BRAF (NM_004333) fusion: c.1115+91797:KCND2_c.1177+1481:BRAFinv Protein Fusion: out of frame {KCND2:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0056952-T01-IM6-138	P-0056952-T01-IM6		Nerve Sheath Tumor	MPNST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493821	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 5248-141:KIAA1549_c.1140+287: BRAFdup Protein Fusion: in frame {KIAA1549: BRAF }	NA
EVT-P-0044270-T02-IM6-139	P-0044270-T02-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491645	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 5247+1560:KIAA1549_c.1140+2463: BRAFdup Protein Fusion: in frame {KIAA1549: BRAF }	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027592-T01-IM6-140	P-0027592-T01-I M6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492343	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c. 5247+769:KIAA1549_c.1140+1765:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0055099-T03-IM6-141	P-0055099-T03-I M6		Miscellaneous Neuroepithelial Tumor	MNET	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490663	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 4929+1357:KIAA1549_c.1141-3279:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0058584-T01-IM7-142	P-0058584-T01-IM7		Nerve Sheath Tumor	NST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491493	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 5247+1663:KIAA1549_c.1140+2615:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0066030-T01-IM7-143	P-0066030-T01-IM7		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482785	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 5248-1172:KIAA1549_c.1314+3615:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035153-T01-IM6-144	P-0035153-T01-IM6		Glioma	GB	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491093	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon15 fused with BRAF exon9) : c.4930-2046:KIAA1549_c.1140+3015:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0035153-T03-IM6-145	P-0035153-T03-IM6		Glioma	GB	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491093	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2046:KIAA1549_c.1140+3015:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0049925-T01-IM6-146	P-0049925-T01-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489497	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 5247+301:KIAA1549_c.1141-2113:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0035844-T03-IM7-147	P-0035844-T03-IM7		Soft Tissue Sarcoma	ERMS	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493285	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 4033-1202:KIAA1549_c.1140+823:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035844-T04-IM7-148	P-0035844-T04-IM7		Soft Tissue Sarcoma	ERMS	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493285	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0017693-T02-IM6-149	P-0017693-T02-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-495:KIAA1549_c.1141-2153:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0062963-T01-IM7-150	P-0062963-T01-IM7		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482198	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 4930-27: KIAA1549_c.1314+623: BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0012575-T01-IM5-151	P-0012575-T01-IM5		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490295	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c. 4929+793: KIAA1549_c.1141-2911: BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052411-T05-IM7-152	P-0052411-T05-IM7		Miscellaneous Neuroepithelial Tumor	MNET	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493530	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0034117-T01-IM6-153	P-0034117-T01-IM6		Breast Cancer	BRCNOS	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140483812	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4346-57:KIAA1549_c.1178-855:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052411-T03-IM6-154	P-0052411-T03-IM6		Miscellaneous Neuroepithelial Tumor	MNET	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493530	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0009097-T01-IM5-155	P-0009097-T01-IM5		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488817	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.5247+3837:KIAA1549_c.1141-1433:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005121-T02-IM5-156	P-0005121-T02-IM5		Melanoma	ACRM	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491248	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c. 4929+2454:KIAA1549_c.1140+2860:BRAF dup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0064872-T01-IM7-157	P-0064872-T01-IM7		Glioma	GB	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489277	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 4032+4347:KIAA1549_c.1141-1893:BRAF dup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013777-T01-IM5-158	P-0013777-T01-IM5		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481674	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549(NM_001164665) - BRAF (NM_004333) rearrangement: c.5248-2734:KIAA1549_c.1315-181:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0040760-T01-IM6-159	P-0040760-T01-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481540	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2627:KIAA1549_c.1315-47: BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014191-T01-IM6-160	P-0014191-T01-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492679	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion : c.5248-320: KIAA1549_c.1140+1429: BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0014381-T01-IM6-161	P-0014381-T01-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492888	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused to BRAF exon 9 -18): c.4930-2082: KIAA1549_c.1140+1220: BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028611-T01-IM6-162	P-0028611-T01-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482786	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015540-T01-IM6-163	P-0015540-T01-IM6		Glioma	GB	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489119	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.4930-69:KIAA1549_c.1141-1735:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0062070-T01-IM7-164	P-0062070-T01-IM7		Glioma	GB	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490443	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1064:KIAA1549_c.1141-3059:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022404-T01-IM6-165	P-0022404-T01-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482177	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 11-18) : c.5248-270:KIAA1549_c.1314+64:BRAF dup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-004032-T01-IM5-166	P-004032-T01-IM5		Glioma	AGNG	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140487892	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon 16 fused to BRAF exon 9) : c.5248-1949:KIAA1549_c.1141-508:BRAF dup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043372-T01-IM6-167	P-0043372-T01-IM6		Miscellaneous Neuroepithelial Tumor	EVN	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140487523	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2204:KIAA1549_c.1141-139:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0028611-T02-IM6-168	P-0028611-T02-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482786	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043874-T01-IM6-169	P-0043874-T01-IM6		Glioma	GNG	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490001	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 5248-2907:KIAA1549_c.1141-2617:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0069024-T04-IM7-170	P-0069024-T04-IM7		Glioma	PAST	KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492117	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 5059:KIAA1549_c.1140+1991:BRAFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029361-T01-IM6-171	P-0029361-T01-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489625	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_00164665) - BRAF (NM_004333) Fusion: c.4930-2667:KIAA1549_c.1141-2241:BRAF dup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0016557-T01-IM6-172	P-0016557-T01-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488576	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_00164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in-frame to BRAF exons 9-18) : c.4929+2955:KIAA1549_c.1141-1192:BRAF dup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032982-T01-IM6-173	P-0032982-T01-I M6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491506	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5248-10 :KIAA1549_c.1140+2602 :BRAFDup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0064310-T04-IM7-174	P-0064310-T04-I M7		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493446	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+3560:KIAA1549_c.1140+662:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0021127-T01-IM6-175	P-0021127-T01-IM6		Breast Cancer	ILC	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491741	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-10 fused with BRAF exons 9-18) : c. 4032+3996:KIAA1549_c.1140+2367:BRAF dup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0017693-T01-IM6-176	P-0017693-T01-IM6		Glioma	GB	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (c. 1141-2153): c.4930-495:KIAA1549_c.1141-2153:BR AFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020125-T01-IM6-177	P-0020125-T01-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492175	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-17 fused to BRAF exons 9-18): c. 5260:KIAA1549_c.1140+1933:BRAFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-0045683-T02-IM6-178	P-0045683-T02-IM6		Glioma	PMA	KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492850	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 5189:KIAA1549_c.1140+1258:BRAFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019127-T01-IM6-179	P-0019127-T01-IM6		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490657	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 9-18): c. 5247+2110:KIAA1549_c.1141-3273:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0066022-T01-IM7-180	P-0066022-T01-IM7		Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488035	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c. 5248-4074:KIAA1549_c.1141-651:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036680-T01-IM6-181	P-0036680-T01-IM6		Melanoma	MUP	KLHL12-BRAF	KLHL12	in-frame	three_prime	140493984	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KLHL12 (NM_021633) - BRAF (NM_004333) fusion (KLHL12 exons 1-4 fused to BRAF exons 9-18): t(1;7)(q32.1;q34)(chr1:g.202880970::chr7:g.140493984) Protein Fusion: in frame {KLHL12:BRAF}	NA
EVT-P-0047833-T02-IM6-183	P-0047833-T02-IM6		Prostate Cancer	PRAD	LRBA-BRAF	LRBA	in-frame	three_prime	140493444	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRBA (NM_001199282) - BRAF (NM_004333) fusion: t(4;7)(q31.3;q34)(chr4:g.151502269::chr7:g.140493444) Protein Fusion: in frame {LRBA:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006412-T01-IM5-184	P-0006412-T01-IM5		Pancreatic Cancer	PAAD	LUC7L2-BRAF	LUC7L2	unknown	five_prime	140501109	6	287	True	286	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		LUC7L2 (NM_001244584) - BRAF (NM_004333) Rearrangement : c.1008-2090:LUC7L2_c.860+103: BRAFdel Antisense fusion	NA
EVT-P-0026158-T04-IM6-185	P-0026158-T04-IM6		Melanoma	SKCM	MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) fusion: c.1417-9863:MAD1L1_c.1140+2359: BRAFdup Protein Fusion: in frame {MAD1L1: BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026158-T01-IM6-186	P-0026158-T01-IM6		Melanoma	MUP	MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) Rearrangement : c.1417-9863_c.1140+2359 dup Protein Fusion: in frame {MAD1L1:BRAF}	NA
EVT-P-0006601-T03-IM6-187	P-0006601-T03-IM6		Breast Cancer	IDC	MINDY4-BRAF	MINDY4	in-frame	three_prime	140490696	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM188B (NM_003222) - BRAF (NM_004333) Fusion (FAM188B exon5 fused with BRAF exon9) : c.664-366:FAM188B_c.1141-3312: BRAF Protein Fusion: in frame {FAM188B:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023220-T01-IM6-188	P-0023220-T01-IM6		Cancer of Unknown Primary	CUP	MKLN1-BRAF	MKLN1	in-frame	three_prime	140496796	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKLN1 (NM_013255) - BRAF (NM_004333) rearrangement: c.98+4070:MKLN1_c.981-2529: BRAFInv Protein Fusion: in frame {MKLN1: BRAF}	NA
EVT-P-0035393-T02-IM6-189	P-0035393-T02-IM6		Thyroid Cancer	THPA	MKRN1-BRAF	MKRN1	in-frame	three_prime	140488436	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.771+833:MKRN1_c.1141-1052: BRAFdup Protein Fusion: in frame {MKRN1: BRAF}	NA
EVT-P-009024-T01-IM5-190	P-009024-T01-IM5		Prostate Cancer	PRAD	MKRN1-BRAF	MKRN1	in-frame	three_prime	140493485	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.772-184G:MKRN1_c.1140+623: BRAFdup Protein fusion: in frame (MKRN1-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008083-T03-IM6-191	P-0008083-T03-IM6		Colorectal Cancer	COAD	MKRN1-BRAF	MKRN1	out-of-frame	three_prime	140484511	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.315-432:MKRN1_c.1178-1554:BRAFdup Protein Fusion: out of frame {MKRN1:BRAF}	NA
EVT-P-001036-T01-IM3-192	P-001036-T01-IM3		Ovarian Cancer	LGSOC	MKRN1-BRAF	MKRN1	in-frame	three_prime	140482250	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-0047135-T01-IM6-193	P-0047135-T01-IM6		Glioma	ASTR	MKRN1-BRAF	MKRN1	in-frame	three_prime	140492510	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-972:MKRN1_c.1140+1598:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004373-T02-IM5-194	P-0004373-T02-IM5		Germ Cell Tumor	MGCT	MKRN1-BRAF	MKRN1	unknown	three_prime	140434466	18	744	False	744	True	exact match	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.314+4351:MKRN1_c.2232:BRAFdup Protein fusion: mid-exon (MKRN1-BRAF)	NA
EVT-P-0009024-T02-IM6-195	P-0009024-T02-IM6		Prostate Cancer	PRAD	MKRN1-BRAF	MKRN1	in-frame	three_prime	140493485	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-185:MKRN1_c.1140+623:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0031896-T02-IM6-196	P-0031896-T02-IM6		Esophageal Cancer	SMN	MKRN1-BRAF	MKRN1	in-frame	three_prime	140482581	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-4 fused to BRAF exons 11-18): c.772-1018:MKRN1_c.1314+240:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-0008083-T02-IM6-197	P-0008083-T02-IM6		Colorectal Cancer	COAD	MKRN1-BRAF	MKRN1	out-of-frame	three_prime	140484505	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-2 fused with BRAF exons 10-18): c.315-428:MKRN1_c.1178-1548:BRAFdup Protein Fusion: out of frame {MKRN1:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013536-T01-IM5-198	P-0013536-T01-IM5		Germ Cell Tumor	SEM	MKRN1-BRAF	MKRN1	in-frame	three_prime	140483618	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.544+172:MKRN1_c.1178-661: BRAFdup Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-0048036-T01-IM6-199	P-0048036-T01-IM6		Melanoma	MEL	NEO1-BRAF	NEO1	in-frame	three_prime	140497651	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NEO1 (NM_002499) - BRAF (NM_004333) fusion: t(7;15)(q34;q24.1)(chr7:g.140497651::chr15:g.73575682) Protein Fusion: in frame {NEO1:BRAF}	NA
EVT-P-0043972-T03-IM6-200	P-0043972-T03-IM6		Non-Small Cell Lung Cancer	LUAD	NRF1-BRAF	NRF1	in-frame	three_prime	140487712	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NRF1 (NM_005011) - BRAF (NM_004333) fusion: c.607-7410:NRF1_c.1141-328:BRAFinv Protein Fusion: in frame {NRF1:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0065446-T01-IM7-201	P-0065446-T01-I-M7		Thyroid Cancer	THPD	NRF1-BRAF	NRF1	in-frame	three_prime	140483810	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NRF1 (NM_005011) - BRAF (NM_004333) fusion: c.1349-10893:NRF1_c.1178-853:BRAFinv Protein Fusion: in frame {NRF1:BRAF}	NA
EVT-P-0059002-T01-IM7-202	P-0059002-T01-I-M7		Prostate Cancer	PRAD	OSBPL7-BRAF	OSBPL7	in-frame	three_prime	140491530	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		OSBPL7 (NM_145798) - BRAF (NM_004333) fusion: t(7;17)(q34;q21.32)(chr7:g.140491530::chr17:g.45890381) Protein Fusion: in frame {OSBPL7:BRAF}	NA
EVT-P-0054958-T02-IM6-203	P-0054958-T02-I-M6		Soft Tissue Sarcoma	MF	PAK2-BRAF	PAK2	in-frame	three_prime	140486065	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PAK2 (NM_002577) - BRAF (NM_004333) rearrangement: t(3;7)(q29;q34)(chr3:g.196530579::chr7:g.140486065) Protein Fusion: in frame {PAK2:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0008789-T01-IM5-205	P-0008789-T01-IM5		Melanoma	SKCM	PHTF2-BRAF	PHTF2	in-frame	three_prime	140489236	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PHTF2 (NM_001127358) - BRAF (NM_004333) rearrangement: c.2224-367:PHTF2_c.1141-1852:BRAF inv Protein fusion: in frame (PHTF2-BRAF)	NA
EVT-P-0044819-T01-IM6-206	P-0044819-T01-IM6		Prostate Cancer	PRAD	PLPP1-BRAF	PLPP1	out-of-frame	three_prime	140482123	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PPAP2A (NM_176895) - BRAF (NM_004333) fusion: t(5;7)(q11.2;q34)(c11p15:g54789192::chr7:g.140482123) Protein Fusion: out of frame {PPAP2A: BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009694-T01-IM5-207	P-0009694-T01-I-M5		Hepatobiliary Cancer	EHCH	PRKAR1B-BRAF	PRKAR1B	in-frame	three_prime	140491855	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRKAR1B (NM_001164758)-BRAF (NM_004333) Rearrangement : c.892-2551:PRKAR1B_c.140+2253:BRAFdup Protein fusion: in frame (PRKAR1B-BRAF)	NA
EVT-P-0008567-T01-IM5-208	P-0008567-T01-I-M5		Thyroid Cancer	THPA	PRKAR2B-BRAF	PRKAR2B	in-frame	three_prime	140486416	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRKAR2B (NM_002736) - BRAF (NM_004333) rearrangement: c.307+8063:PRKAR2B_c.1177+932:BRAFInv Protein fusion: in frame (PRKAR2B-BRAF)	NA
EVT-P-0041181-T01-IM6-209	P-0041181-T01-I-M6		CNS Cancer	SUBE	PTPRZ1-BRAF	PTPRZ1	in-frame	three_prime	140492428	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PTPRZ1 (NM_002851) - BRAF (NM_004333) fusion: c.5367+457:PTPRZ1_c.1140+1680:BRAFInv Protein Fusion: in frame {PTPRZ1:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012884-T01-IM5-210	P-0012884-T01-IM5		Melanoma	SKCM	PWWP2A-BRAF	PWWP2A	in-frame	three_prime	140495243	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PWWP2A (NM_001130864) - BRAF(NM_004333) Rearrangement: t(5;7)(q33;q34)(chr5:g.159539324::chr7:g.140495243) Protein fusion: in frame (PWWP2A-BRAF)	NA
EVT-P-0068260-T01-IM7-211	P-0068260-T01-IM7		Non-Small Cell Lung Cancer	LUAD	RBMS3-BRAF	RBMS3	in-frame	three_prime	140482707	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RBMS3 (NM_001003793) - BRAF (NM_004333) fusion: t(3;7)(p24.1;q34)(chr3:g.30030473::chr7:g.140482707) Protein Fusion: in frame {RBMS3: BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064764-T01-IM7-212	P-0064764-T01-I-M7		Pancreatic Cancer	PAAC	RGS3-BRAF	RGS3	out-of-frame	three_prime	140493282	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RGS3 (NM_144488) - BRAF (NM_004333) fusion: t(7;9)(q34;q32)(chr7:g.140493282::chr9:g.116291333) Protein Fusion: out of frame {RGS3:BRAF}	NA
EVT-P-0041308-T02-IM6-213	P-0041308-T02-I-M6		Colorectal Cancer	COAD	RRBP1-BRAF	RRBP1	in-frame	three_prime	140485917	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RRBP1 (NM_004587) - BRAF (NM_004333) fusion: t(7;20)(q34;p12.1)(chr7:g.140485917::chr20:g.17634546) Protein Fusion: in frame {RRBP1:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007392-T01-IM5-214	P-0007392-T01-IM5		Thyroid Cancer	THPD	SCRIB-BRAF	SCRIB	in-frame	three_prime	140482150	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SCRIB (NM_182706) - BRAF (NM_004333) rearrangement: t(7;8)(q34;q24.3)(chr7:g.140482150::chr8:g.144876675) Protein fusion: in frame (SCRIB-BRAF)	NA
EVT-P-0056639-T01-IM6-215	P-0056639-T01-IM6		Hepatobiliary Cancer	IHCH	SETBP1-BRAF	SETBP1	in-frame	three_prime	140486594	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SETBP1 (NM_015559) - BRAF (NM_004333) fusion: t(7;18)(q34;q12.3)(chr7:g.140486594::chr18:g.42638299) Protein Fusion: in frame {SETBP1:BRAF}	NA
EVT-P-0048950-T01-IM6-216	P-0048950-T01-IM6		Prostate Cancer	PRAD	SND1-BRAF	SND1	in-frame	three_prime	140492033	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+30037:SND1_c.1140+2075: BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012395-T01-IM5-217	P-0012395-T01-I M5		Pancreatic Cancer	PAAD	SND1-BRAF	SND1	in-frame	three_prime	140493527	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1152+27174:SND1_c.1140+581:BRAF inv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0009619-T01-IM5-218	P-0009619-T01-I M5		Pancreatic Cancer	PAAD	SND1-BRAF	SND1	in-frame	three_prime	140482494	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1968+2394:SND1_c.1314+327:BRAF inv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0052299-T04-IM6-219	P-0052299-T04-I M6		Prostate Cancer	PRAD	SND1-BRAF	SND1	unknown	three_prime	140482935	10	400	False	400	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1038+2929:SND1_c.1200: BRAF inv Protein Fusion: mid-exon {SND1:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001687-T01-IM3-220	P-0001687-T01-I M3		Prostate Cancer	PRSCC	SND1-BRAF	SND1	in-frame	three_prime	140490224	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) inversion: c.1152+28381_c.1141-2840inv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0004103-T01-IM5-221	P-0004103-T01-I M5		Non-Small Cell Lung Cancer	LUAD	SND1-BRAF	SND1	in-frame	three_prime	140481685	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1038+5799: SND1_c.1315-192: BRAF inv. Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0029727-T01-IM6-222	P-0029727-T01-I M6		Thyroid Cancer	THPA	SND1-BRAF	SND1	in-frame	three_prime	140482257	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+12095: SND1_c.1314+564: BRAF inv Protein Fusion: in frame {SND1: BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068194-T01-IM7-223	P-0068194-T01-IM7		Pancreatic Cancer	PAAD	SND1-BRAF	SND1	in-frame	three_prime	140489175	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+17686:SND1_c.1141-1791:BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0008140-T01-IM5-224	P-0008140-T01-IM5		Pancreatic Cancer	PAAC	SND1-BRAF	SND1	in-frame	three_prime	140489439	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 fused with BRAF exons 9-18) : c.1039-3849:SND1_c.1141-2055:BRAFinv Protein fusion: in frame (SND1-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034168-T01-IM6-225	P-0034168-T01-IM6		Melanoma	SKCM	SND1-BRAF	SND1	in-frame	three_prime	140489421	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused to BRAF exons 9-18): c. 1152+6972:SND1_c.1141-2037:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0025075-T01-IM6-226	P-0025075-T01-IM6		Ampullary Cancer	PAMPC A	SND1-BRAF	SND1	in-frame	three_prime	140490546	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 with BRAF exons 9-18) : c. 1039-2471:SND1_c.1141-3162:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002071-T02-IM5-227	P-0002071-T02-IM5		Pancreatic Cancer	PAAD	SND1-BRAF	SND1	in-frame	three_prime	1404892	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused with BRAF exons 9-18) : c.1152+18606:SND1_c.1141-1908:BRAFInv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0002197-T01-IM3-228	P-0002197-T01-IM3		Pancreatic Cancer	PAAC	SND1-BRAF	SND1	in-frame	five_prime	1404883	8	380	True	380	True	exact match	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		c.1528-8436:SND1_c.1141-922:BRAFInv Protein fusion: in frame (BRAF-SND1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007533-T01-IM5-229	P-0007533-T01-I M5		Prostate Cancer	PRAD	SND1-BRAF	SND1	in-frame	three_prime	14048226	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738:SN D1_c.1314+595 :BRAFinv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0036567-T01-IM6-230	P-0036567-T01-I M6		Prostate Cancer	PRAD	SND1-BRAF	SND1	in-frame	three_prime	140493552	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) Fusion (SND1 exon10 fused with BRAF exon9) : c.1153-22013:SN D1_c.1140+556: BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034845-T01-IM6-231	P-0034845-T01-IM6		Prostate Cancer	PRAD	SND1-BRAF	SND1	in-frame	three_prime	140489241	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1527+3658: SND1_c.1141-1857:BRAF inv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0054364-T01-IM6-232	P-0054364-T01-IM6		Hepatobiliary Cancer	IHCH	SNX8-BRAF	SNX8	in-frame	three_prime	140488730	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SNX8 (NM_013321) - BRAF (NM_004333) fusion: c.1284+565:SNX8_c.1141-1346:BRAF Fdup Protein Fusion: in frame {SNX8:BRAF}	NA
EVT-P-0026773-T01-IM6-234	P-0026773-T01-IM6		Pancreatic Cancer	PAAD	SVOPL-BRAF	SVOPL	unknown	five_prime	140482097	10	438	True	438	True	exact match	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) rearrangement: c.1315-604:BRAF_chr7:g.138277426del Transcript Fusion {BRAF:SVOPL}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051214-T01-IM6-235	P-0051214-T01-IM6		Prostate Cancer	PRAD	TMPRS S2-BRAF	TMPRS S2	in-frame	three_prime	140502592	6	238	True	238	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain	Cysteine-rich domain	TMPRS S2 (NM_001135099) - BRAF (NM_004333) fusion: t(7;21)(q34;q22.3)(chr7:g.140502592::chr21:g.42867846) Protein Fusion: in frame {TMPRS S2:BRAF}	NA
EVT-P-0018524-T01-IM6-236	P-0018524-T01-IM6		Prostate Cancer	PRAD	TMPRS S2-BRAF	TMPRS S2	out-of-frame	three_prime	140490337	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TMPRS S2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(21;7)(q22.3;q34)(chr21:g.42873801::chr7:g.140490337) Protein Fusion: out of frame {TMPRS S2:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022778-T01-IM6-237	P-0022778-T01-IM6		Prostate Cancer	PRAD	TMPRS S2-BRAF	TMPRS S2	in-frame	three_prime	140483147	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TMPRS S2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(7;21)(q34;q22.3)(chr7:g.140483147::chr21:g.42878796) Protein Fusion: in frame {TMPRS S2:BRAF}	NA
EVT-P-0043836-T01-IM6-239	P-0043836-T01-IM6		Pancreatic Cancer	PAAC	TRA2B-BRAF	TRA2B	in-frame	three_prime	140491978	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRA2B (NM_004593) - BRAF (NM_004333) fusion: t(3;7)(q27.2;q34)(chr3:g.185650761::chr7:g.140491978) Protein Fusion: in frame {TRA2B:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054430-T02-IM6-240	P-0054430-T02-IM6		Colorectal Cancer	COAD	TRIM24-BRAF	TRIM24	in-frame	three_prime	140487875	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c. 2719-489:TRIM24_c.1141-491:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0025699-T02-IM6-241	P-0025699-T02-IM6		Non-Small Cell Lung Cancer	LUAD	TRIM24-BRAF	TRIM24	in-frame	three_prime	140486854	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c. 2014+674:TRIM24_c.1177+494:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0045111-T01-IM6-242	P-0045111-T01-IM6		Colorectal Cancer	COAD	TRIM24-BRAF	TRIM24	in-frame	three_prime	140488049	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c. 2257-760:TRIM24_c.1141-665:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA

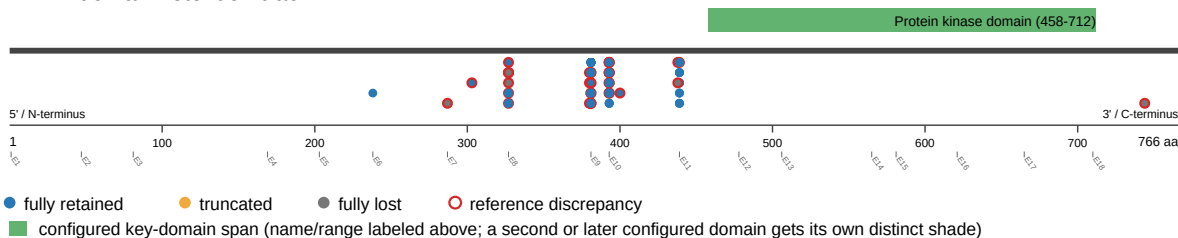
event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0066835-T01-IM7-243	P-0066835-T01-I M7		Colorectal Cancer	COAD	TRIM24-BRAF	TRIM24	in-frame	three_prime	140490589	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c. 1879-502:TRIM24_c.1141-3205:BRAFinv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0068451-T01-IM7-244	P-0068451-T01-I M7		Cancer of Unknown Primary	ADNOS	TRIM24-BRAF	TRIM24	in-frame	three_prime	140482765	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c. 484-860:TRIM24_c.1314+56:BRAFinv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0030988-T01-IM6-245	P-0030988-T01-I M6		Colorectal Cancer	COAD	TRIM24-BRAF	TRIM24	unknown	three_prime	14049102	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c. 2349:TRIM24_c.1140+3106:BRAFinv Protein Fusion: mid-exon {TRIM24:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032569-T01-IM6-246	P-0032569-T01-IM6		Colorectal Cancer	COAD	TRIM24-BRAF	TRIM24	in-frame	three_prime	140487936	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) Rearrangement : c.1878+791:TRIM24_c.1141-552inv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0032569-T02-IM6-247	P-0032569-T02-IM6		Colorectal Cancer	COAD	TRIM24-BRAF	TRIM24	in-frame	three_prime	140487936	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1878+791:TRIM24_c.1141-552:BRAFinv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0019039-T01-IM6-248	P-0019039-T01-IM6		Glioma	ASTR	UBE2H-BRAF	UBE2H	out-of-frame	three_prime	140499248	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) - UBE2H (NM_003344) rearrangement: c.980+914:BR AF_c.299-4611:UBE2Hdup Protein Fusion: out of frame {UBE2H:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027762-T01-IM6-250	P-0027762-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ZC3HAV1-BRAF	ZC3HAV1	in-frame	three_prime	140485946	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZC3HAV1 (NM_020119) - BRAF (NM_004333) rearrangement: c.698-431:ZC3HAV1_c.1177+1402:BRAFdup Protein Fusion: in frame {ZC3HAV1:BRAF}	NA
EVT-P-0008920-T01-IM5-251	P-0008920-T01-IM5		Prostate Cancer	PRAD	ZNF207-BRAF	ZNF207	in-frame	three_prime	140485847	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZNF207 (NM_001098507) - BRAF (NM_004333) rearrangement: t(7;17)(q34;q11.2)(c17:g.140485847::chr17:g.30687533) Protein fusion: in frame (ZNF207-BRAF)	NA

domain retention outliers

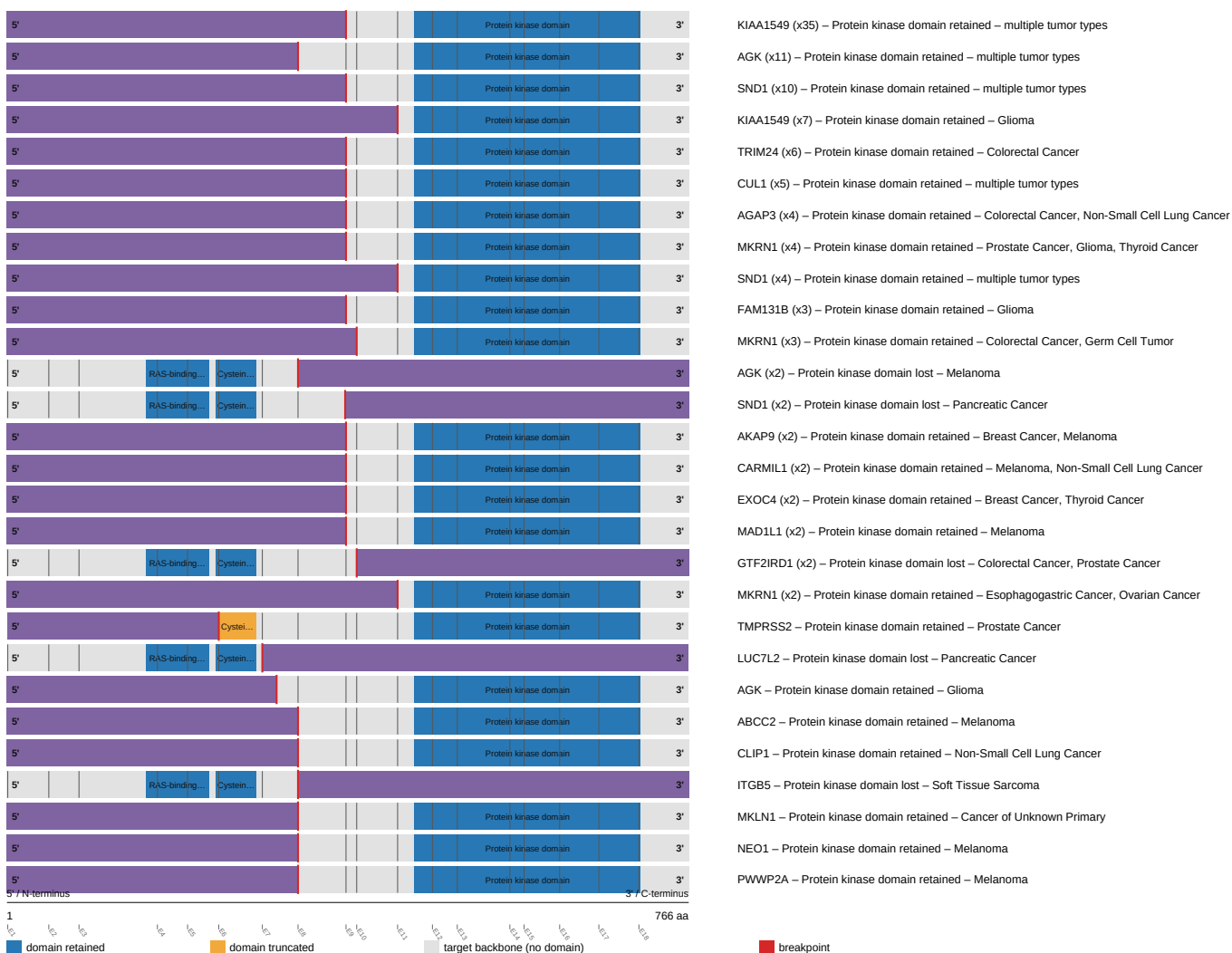
BRAF domain-retention track



fusion schematic

BRAF fusion-transcript schematic

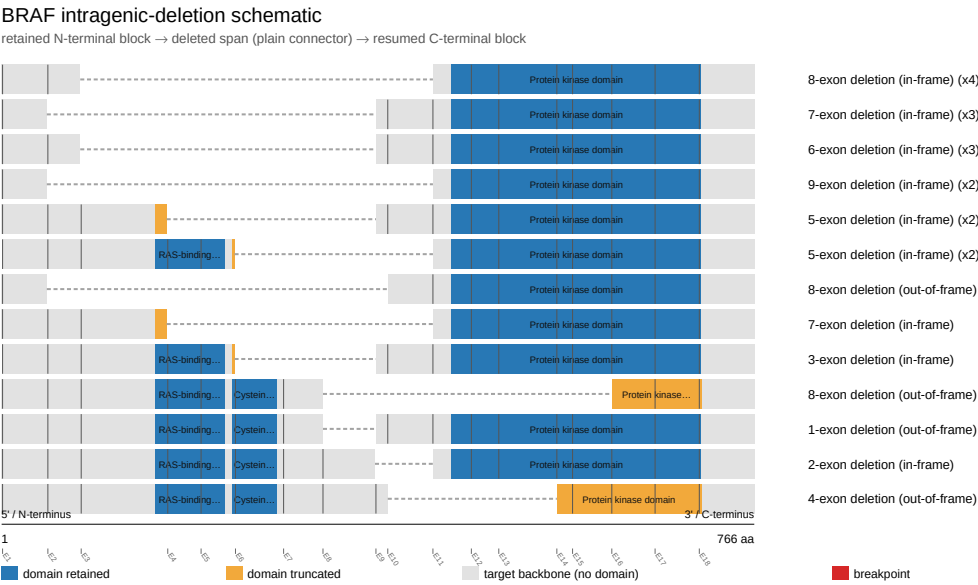
Fusion order varies by row; follow each row's 5' and 3' labels (BRAF portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 89 partner/breakpoint groups by recurrence: see results.tsv for the complete list.

intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

